



## Course Content of java [Total 95 sessions]

| Index | Topics                                                                                                                            | Duration |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| 1     | Introduction to Language                                                                                                          | 2 Hours  |
| 2     | Introduction to Java Language                                                                                                     | 4 Hours  |
|       | a) Flavors of Java, History of Java, Features of Java, Introduction to JDK, JRE, JVM and JIT Compiler.                            |          |
| 3     | Platform Independency in java                                                                                                     | 2 Hours  |
|       | a) Difference between Compiler and Interpreter                                                                                    |          |
| 4     | Token in Java                                                                                                                     | 2 Hours  |
|       | a) Keywords, Identifiers, Literals, Punctuators and Operators                                                                     |          |
| 5     | Moving towards First Program of Java                                                                                              | 4 Hours  |
|       | a) Description of main method                                                                                                     |          |
|       | b) How to download and install Java                                                                                               |          |
|       | c) First Java program using Notepad, Edit plus and Eclipse IDE                                                                    |          |
| 6     | Types of Literals in java                                                                                                         | 6 Hours  |
|       | a) Integral Literal<br>b) Floating point Literal<br>c) Char Literal<br>d) Boolean Literal<br>e) String Literal<br>f) null literal |          |
| 7     | Operators                                                                                                                         | 2 Hours  |
|       | a) Arithmetic Operator<br>b) Unary Operators<br>c) Assignment Operator                                                            |          |

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|    | d) Relational Operator<br>e) Logical Operators<br>f) Boolean Operators<br>g) Bitwise Operators<br>h) Ternary Operator<br>i) Member Operator(.)<br>j) new Operator<br>k) instanceof Operator |          |
| 9  | Programs on Method Parameter and return type                                                                                                                                                | 2 Hours  |
| 10 | Introduction to Object Oriented Programming                                                                                                                                                 | 20 Hours |
|    | a) OOPs Features and Advantages                                                                                                                                                             |          |
|    | b) Class, Object, Abstraction, Encapsulation, Inheritance and Polymorphism                                                                                                                  |          |
|    | c) Default constructor added by compiler                                                                                                                                                    |          |
|    | d) Why compiler adds default constructor to our class                                                                                                                                       |          |
|    | e) Types of variables (Primitive and Reference)                                                                                                                                             |          |
|    | f) Instance variable, Static variable, Parameter variable and local variable                                                                                                                |          |
|    | g) How to provide our own user defined values for instance variable                                                                                                                         |          |
|    | h) this keyword                                                                                                                                                                             |          |
|    | i) Role of instance variable while creating the Object                                                                                                                                      |          |
|    | j) Working with static variable while creating the Object                                                                                                                                   |          |
|    | k) When we should declare a variable as an instance or static variable?                                                                                                                     |          |
|    | l) Data Hiding                                                                                                                                                                              |          |
|    | m) Abstraction                                                                                                                                                                              |          |
|    | n) Encapsulation                                                                                                                                                                            |          |
|    | o) How to print object properties value (instance variable value)                                                                                                                           |          |

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|    | p) Setter and Getter                                                    |          |
| 10 | Introduction to Constructor                                             | 4 Hours  |
|    | a) Advantage of Constructor                                             |          |
|    | b) Types Of Constructor                                                 |          |
|    | c) Default, No Argument and Parameterized Constructor                   |          |
|    | d) Passing Object reference to the constructor(Copy Constructor)        |          |
|    | e) Instance Block in java                                               |          |
|    | f) How many ways we can initialize object properties(instance variable) |          |
| 11 | Relationship between the classes                                        | 8 Hours  |
|    | a) IS-A (Inheritance) Relation and HAS-A(Association) Relation          |          |
|    | b) Introduction to Inheritance (IS-A relation)                          |          |
|    | c) Types of Inheritance                                                 |          |
|    | d) this() and super()                                                   |          |
|    | e) Why java does not support multiple inheritance                       |          |
|    | f) Access modifiers in Java                                             |          |
|    | g) HAS-A relation(Association)                                          |          |
|    | h) Composition and Aggregation                                          |          |
| 12 | Wrapper classes in Java                                                 | 2 Hours  |
|    | a) Autoboxing and Unboxing                                              |          |
| 13 | Introduction to Polymorphism                                            | 10 Hours |
|    | a) Method Overloading, Var-Args,                                        |          |
|    | b) Ambiguity issues while overloading a method                          |          |
|    | c) Method Overriding                                                    |          |

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|    | d) Upcasting and Downcasting                                                                                                                        |          |
|    | e) @Override Annotation                                                                                                                             |          |
|    | f) Role of Access Modifier while Overriding a method                                                                                                |          |
|    | g) Co-variant concept in method overriding                                                                                                          |          |
|    | h) Method Hiding                                                                                                                                    |          |
| 14 | final keyword in Java                                                                                                                               | 2 Hours  |
| 15 | Object class and its method                                                                                                                         | 4 Hours  |
|    | a) getClass(), hashCode(), toString(), equals(Object o)                                                                                             |          |
| 16 | Inner classes in java                                                                                                                               | 4 Hours  |
|    | a) Nested inner class, Method local inner class, static nested inner class, Anonymous inner class                                                   |          |
| 17 | Abstract class and abstract methods                                                                                                                 | 4 Hours  |
| 18 | Introduction to interface                                                                                                                           | 10 Hours |
|    | a) Default and static method( <b>Java 8 features</b> )                                                                                              |          |
|    | b) Functional interfaces                                                                                                                            |          |
|    | c) Lambda Expression                                                                                                                                |          |
|    | d) Predicate<T>, Consumer<T>, Supplier<T>, Function<T,R>, BiPredicate<T,U>, BiFunction<T,U>, BiFunction<T,U,R>, UnaryOperator<T>, BinaryOperator<T> |          |
| 19 | Enum in java                                                                                                                                        | 2 Hours  |

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| 20 | JVM Architecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 8 Hours  |
|    | a) Class loader subsystem, Runtime Data areas and Execution Engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|    | b) Different types of class loaders                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
|    | c) Method Area, Heap Memory, Stack Memory, PC register, Native Method Stack.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
|    | d) Execution Engine and JIT Compiler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| 21 | Arrays in java                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 Hours |
|    | a) Array Introduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
|    | b) Characteristics of an Array                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
|    | c) Drawback of an Array                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|    | d) Types of Arrays                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|    | Programs:<br>1) WAP to find first and last index of Array<br>2) WAP to print Array elements in reverse<br>3) WAP to find sum of all the elements<br>4) Calculate Student marks in find average<br>5) Working with Custom Object<br>6) Storing Book object to access Book property<br>7) Create Animal abstract class with sub classes show using dynamic Polymorphism.<br>8) Create Vehicle interface and sub classes to sh dynamic polymorphism.<br>9) Object array by using Literal.<br>10) Arrays class Methods.<br>11) Remaining Logical By using <b>Stream API</b> |          |
| 22 | String Handling in Java                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10 Hours |
|    | a) String Immutability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|    | b) Working with Various Methods of String class like length(), charAt(), equals(Object obj),                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |

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|    | equalsIgnoreCase(), toUpperCase(), toLowerCase(), replace(), trim(), indexOf(), lastIndexOf(), split(), toCharArray(), getBytes() e.t.c |          |
|    | c) == operator and equals(Object obj) method                                                                                            |          |
|    | d) StringBuffer class and its method                                                                                                    |          |
|    | e) StringBuilder class and its method                                                                                                   |          |
|    | g) Difference between String, StringBuffer and StringBuilder.<br>h) Performance Comparison between StringBuffer and StringBuilder.      |          |
| 23 | Exception Handling in Java                                                                                                              | 12 Hours |
|    | a) try-catch block                                                                                                                      |          |
|    | b) Nested try catch, try with multi catch block                                                                                         |          |
|    | c) Dealing with Infinity and NaN                                                                                                        |          |
|    | d) Finally block, try with resources                                                                                                    |          |
|    | e) Exception Propagation                                                                                                                |          |
|    | f) Checked and Unchecked Exception                                                                                                      |          |
|    | g) throw and throws keyword                                                                                                             |          |
|    | h) User-defined checked and unchecked Exception                                                                                         |          |
|    | i) Rules related to checked Exception                                                                                                   |          |
| 23 | Introduction to Multithreading                                                                                                          | 14 Hours |

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|    | a) Process and Thread, Multitasking and multithreading                    |          |
|    | b) Creating Thread by using Thread class and Runnable interface           |          |
|    | c) Various methods of Thread class                                        |          |
|    | d) Implementation of Runnable interface by using <b>Lambda Expression</b> |          |
|    | e) Race condition in multithreading                                       |          |
|    | f) Synchronization (Method and block level)                               |          |
|    | g) Object and class level                                                 |          |
|    | Synchronization                                                           |          |
|    | h) Thread life cycle                                                      |          |
|    | i) Thread Group and Volatile                                              |          |
|    | j) Inter Thread Communication(ITC)                                        |          |
|    | k) Deadlock in multithreading                                             |          |
|    | l) interrupt(), Daemon Thread                                             |          |
| 24 | Introduction to Java I/O Streams                                          | 2 Hours  |
|    | a) Working with Binary Stream and Character Stream                        |          |
|    | b) Serialization and De-Serialization                                     |          |
|    | c) transient keyword role in Serialization                                |          |
| 25 | Collection Framework                                                      | 26 Hours |
|    | a) Collection Hierarchy                                                   |          |

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|  | b) List,Set and Queue interface                                                                                                                  |  |
|  | c) 9 ways to retrieve the object                                                                                                                 |  |
|  | from Collection (Including<br>forEach() and Method<br>Reference)                                                                                 |  |
|  | d) Working with Custom<br>Object(ArrayList<Product>)                                                                                             |  |
|  | e) Implementation of List<br>interface classes (ArrayList,<br>LinkedList, Vector and Stack)                                                      |  |
|  | f) Hashing technique                                                                                                                             |  |
|  | g) Implementation of Set<br>interface classes (HashSet,<br>LinkedHashSet)                                                                        |  |
|  | h) SortedSet and NavigableSet<br>interface                                                                                                       |  |
|  | i) Comparable and Comparator<br>interface                                                                                                        |  |
|  | j) TreeSet class and its<br>implementation                                                                                                       |  |
|  | k) Introduction to Map interface                                                                                                                 |  |
|  | l) Entry interface                                                                                                                               |  |
|  | m)Implementation of Map<br>interface classes (Hashtable,<br>HashMap,<br>ConcurrentHashMap,<br>LinkedHashMap,<br>WeakHashMap,<br>IdentityHashMap) |  |
|  | n) Properties class                                                                                                                              |  |

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|    | o) SortedMap and NavigableMap                                                                                                                                                              |                             |
|    | p) TreeMap class and its implementation                                                                                                                                                    |                             |
|    | q) Queue interface, PriorityQueue                                                                                                                                                          |                             |
| 26 | Working with Generics                                                                                                                                                                      | 4 Hours                     |
|    | Mixing generic and non-generic collections                                                                                                                                                 |                             |
|    | Polymorphism with Generic                                                                                                                                                                  |                             |
|    | Type Erasure in Generic                                                                                                                                                                    |                             |
|    | Wild Card in Generic<?>                                                                                                                                                                    |                             |
| 27 | Streams API                                                                                                                                                                                | 8 hours                     |
|    | Creation of Streams to process the data                                                                                                                                                    |                             |
|    | Working with intermediate and Terminal methods (22 Methods).<br>Finite and Infinite Stream.<br><br><b>Special Logical Array and String Programs Using STREAM API.<br/>[Placement Team]</b> |                             |
| 28 | New Features of Java                                                                                                                                                                       | 4 Hours                     |
|    | Date API, Optional class, Record class, Sealed class in java                                                                                                                               |                             |
| 29 | Executor Framework                                                                                                                                                                         | <b>Video using Workshop</b> |
|    | a) Executor<br>b) ExecutorService<br>c) ScheduledExecutorService<br>d) Advantages of Using the Executor Framework                                                                          |                             |